Unit 4·4 Indeterminate forms and L'Hospitals rule p 301-309

Note Title 2014/04/09

Previously we could do;

① $\lim_{x \to 3} \frac{x-3}{x^2-9}$ $\left(\frac{0}{0}\right)$

$= \lim_{x \to 3} \frac{\cancel{x-3}}{(x+3)\cancel{(x-3)}} = \frac{1}{6}$

② $\lim_{x \to \infty} \frac{x^5+5x^3}{x^4+x^2}$ $\left(\frac{\infty}{\infty}\right)$

$= \infty$

What about?

$\lim_{x \to 1} \frac{\ln x}{x-1}$ $\left(\frac{0}{0}\right)$??

Indeterminate forms: (7 such forms)

$\frac{0}{0}$ ✓, $\frac{\pm\infty}{\pm\infty}$ ✓ $\infty - \infty$ ✓ $\underline{\underline{0 \times \infty}}$

0^0, ∞^0, 1^∞

[0^∞ - this is not indeterminate] → Ex 86 p 314

$\boxed{\text{L'Hospital's Rule:}}$ p 302

If $\lim\limits_{x \to \infty} \dfrac{f(x)}{g(x)}$ is of the form $\left(\frac{0}{0}\right)$ or $\left(\frac{\pm\infty}{\pm\infty}\right)$

then $$\boxed{\lim_{x \to \infty} \frac{f(x)}{g(x)} = \lim_{x \to \infty} \frac{f'(x)}{g'(x)}}$$

This is also true for $\lim\limits_{x \to -\infty}$ or $\lim\limits_{x \to a}$ and left and right limits.

Examples:

(1) $$\lim_{x \to \infty} \frac{3^x}{x^2} \quad \left(\frac{\infty}{\infty}\right)$$

$$\stackrel{H}{=} \lim_{x \to \infty} \frac{3^x \ln 3}{2x} \quad \left(\frac{\infty}{\infty}\right)$$

$$\stackrel{H}{=} \lim_{x \to \infty} \frac{3^x (\ln 3)^2}{2} = \infty$$

$$\lim_{x \to 1} \frac{\ln x}{x-1} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \to 1} \frac{\frac{1}{x}}{1} = 1$$

(2) $$\lim_{x \to -\infty} \frac{3^x}{x^2} \quad \left(\frac{0}{\infty}\right)$$

$$= 0$$

(3) $$\lim_{x \to \infty} \frac{\ln x}{x^2} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \to \infty} \frac{\frac{1}{x}}{2x} = \lim_{x \to \infty} \frac{1}{2x^2} = 0$$

(4) $$\lim_{x \to -\infty} \frac{\ln x}{x^2}$$

dne.

$(x > 0)$

(5) $\lim_{x\to 0} \frac{2x}{e^{3x}-1}$ $\left(\frac{0}{0}\right)$

$= \lim_{x\to 0} \frac{2}{e^{3x}\,3} = \frac{2}{3}$

(6) $\lim_{x\to \pi/2} \frac{\cos x}{2x-\pi}$ $\left(\frac{0}{0}\right)$

$= \lim_{x\to \pi/2} \frac{-\sin x}{2}$

$= -1/2$

(7) $\lim_{x\to\infty} \frac{x\ln x}{x+\ln x}$

$= \lim_{x\to\infty} \frac{\ln x + \cancel{x}\cdot\frac{1}{\cancel{x}}}{1+\cancel{\frac{1}{x}}\ 0}$

$= \infty$

$\boxed{\infty - \infty}$

1. $\lim_{x\to\infty} (x^3 - x^4)$ $(\infty - \infty)$

$= \lim_{x\to\infty} x^3(1-x)$ $(\infty \cdot -\infty)$ not an interterminate form.

$= -\infty$

2. $\lim_{x\to\infty} (x^3 + x^4)$ $(\infty + \infty)$

$= \infty$

3. $\lim_{x\to -\infty} (x^4 + x^3)$ $(\infty - \infty)$

$= \lim_{x\to -\infty} x^3(x+1)$ $(-\infty \cdot -\infty)$

$= \infty$

4. $\displaystyle\lim_{x\to 1^+}\left(\frac{1}{\ln x}-\frac{1}{x-1}\right)$ $\quad(\infty-\infty)$

$$=\lim_{x\to 1^+}\left(\frac{(x-1)-\ln x}{\ln x\,(x-1)}\right)\quad\left(\frac{0}{0}\right)$$

$$=\lim_{x\to 1^+}\left(\frac{1-\frac{1}{x}}{\frac{1}{x}(x-1)+\ln x}\right)\begin{matrix}\times x\\ \times x\end{matrix}\quad\left(\frac{0}{0}\right)$$

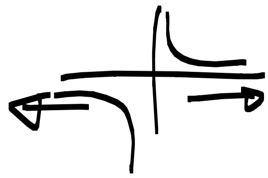
$$=\lim_{x\to 1^+}\left(\frac{1}{1+\ln x+1}\right)\quad[\text{Check!!!}]$$

$$=\frac{1}{2}$$

$\boxed{\infty\times 0}$

1. $\displaystyle\lim_{x\to -\infty} x e^{x}$ $\quad(\underline{\underline{-\infty\times 0}})$

$$=\lim_{x\to -\infty}\frac{x}{e^{-x}}\quad\left(\frac{-\infty}{\infty}\right)$$



$\frac{1}{x}$

$$=\lim_{x\to -\infty}\frac{1}{-e^{-x}}\quad=0$$

2. $\displaystyle\lim_{x\to \infty} x.e^{3/x}$ $\quad(\infty\times 1)$

$$=\infty$$

We have now dealt with

$\frac{0}{0}$ ✓, $\frac{\pm\infty}{\pm\infty}$ ✓, $\infty - \infty$ ✓, $0 \times \infty$ ✓ * L'Hosp

What about $\boxed{0^0, \quad \infty^0 \quad \text{and} \quad 1^\infty}$

<u>Examples</u>:

1. $\lim_{x \to \infty} \left(1 + \frac{4}{x}\right)^{3x}$ $\quad (1^\infty)$

Let $y = \left(1 + \frac{4}{x}\right)^{3x}$

$$\Rightarrow \ln y = 3x \ln\left(1 + \frac{4}{x}\right)$$

$$\Rightarrow \lim_{x \to \infty} \ln y = \lim_{x \to \infty} 3x \cdot \ln\left(1 + \frac{4}{x}\right) \quad (\infty \cdot 0)$$

$$= 3 \lim_{x \to \infty} \frac{\ln\left(1 + \frac{4}{x}\right)}{\frac{1}{x}} \quad \left(\frac{0}{0}\right)$$

$$\overset{H}{=} 3 \lim_{x \to \infty} \frac{\left.\frac{-\frac{4}{x^2}}{1 + \frac{4}{x}}\right\} \times x^2}{\left.-\frac{1}{x^2}\right\} \times x^2}$$

$$= 3 \lim_{x \to \infty} \frac{\frac{-4}{1 + \frac{4}{x}}}{-1} = 12 \lim_{x \to \infty} \frac{1}{1 + \frac{4}{x}}$$

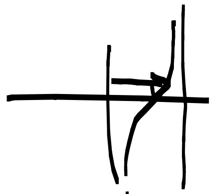
$$= 12 \rightarrow \quad * \quad *$$

Oops!!

So $\lim_{x \to \infty} \ln y = 12$ (Theorem 8 p120.)

$$\Rightarrow \ln \lim_{x\to\infty} y = 12 \quad **$$

$$\Rightarrow \lim_{x\to\infty} y = e^{12}$$



tanx.

1. $\lim_{x \to \pi/2^-} (\sec x - \tan x)$ $(\infty - \infty)$

$$= \lim_{x\to\pi/2^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right)$$

$$= \lim_{x\to\pi/2^-} \left(\frac{1-\sin x}{\cos x} \right) \quad \left(\frac{0}{0}\right)$$

$$\overset{H}{=} \lim_{x\to\pi/2^-} \left(\frac{+\cos x}{+\sin x} \right) = 0$$

2. $\lim_{x\to 0} \frac{\cos x}{x^2}$ $\left(\frac{c}{0}\right)$

$$= \infty$$

3. $\lim\limits_{x\to\infty} (e^x+6x)^{2/x}$ $\qquad (\infty)^0$

Let $y = (e^x+6x)^{2/x}$

$\Rightarrow \ln y = \frac{2}{x}\cdot\ln(e^x+6x)$ $\qquad (0\times\infty)$

$$\lim_{x\to\infty}\ln y = 2\lim_{x\to\infty}\frac{\ln(e^x+6x)}{x} \qquad \left(\frac{\infty}{\infty}\right)$$

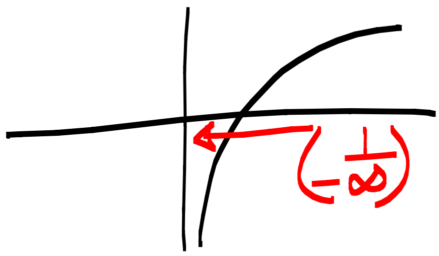
$$= 2\lim_{x\to\infty}\frac{e^x+6}{e^x+6x} \qquad \left(\frac{\infty}{\infty}\right)$$

$$= 2\lim_{x\to\infty}\frac{e^x}{e^x+6} \qquad \left(\frac{\infty}{\infty}\right)$$

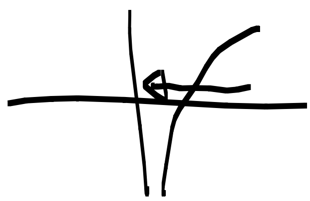
$$= 2\lim_{x\to\infty}\frac{\cancel{e^x}}{\cancel{e^x}} = 2$$

$$\ln \lim_{x\to\infty} y = 2$$

$$\lim_{x\to\infty} y = e^2$$



4. $\lim\limits_{x\to0^+} 2x^{\frac{1}{\ln x}}$ $\qquad (0^0)$



Let $y = 2x^{\frac{1}{\ln x}}$

$$\ln y = \frac{1}{\ln x}\cdot\ln 2x$$

$$\lim_{x\to0^+}\ln y = \lim_{x\to0^+}\frac{\ln 2x}{\ln x} \qquad \left(\frac{-\infty}{-\infty}\right)$$

$$= \lim_{x \to 0^+} \frac{\frac{2}{2x}}{\frac{1}{x}} = 1$$

$$\lim_{x \to 0^+} y = e^1 = e$$

What can I try when determining indeterminate limits?

In a perfect world:

- $\left(\frac{0}{0}\right)$ and $\left(\frac{\pm\infty}{\pm\infty}\right)$ – try L'Hospital.
- Rational functions – use technique of rational functions.
- $\infty - \infty$ – try factorizing
- $0 \times \infty$ – write as a fraction and then try L'Hospital.
- 0^0, ∞^0 or 1^∞ – set function $= y$
  Introduce ln
  write as fraction and try L'Hospital.

WTW 238 Prove that!

4. $$\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e \;!!$$

Examples:

1. $\lim_{x \to \pi/2^-} (\sin x)^{\tan^2 x}$ $\left(e^{-1/2}\right)$

2. $\lim_{x \to 0^+} \left(\frac{1}{x}\right)^{\sin x}$ (1)

3. $\lim_{x \to \infty} x\left(e^{\frac{3}{x}} - 1\right)$ (-3)

1. Let $y = (\sin x)^{\tan^2 x}$ $\quad (1^\infty)$

$\Rightarrow \ln y = \underbrace{\tan^2 x}_{\infty} \ln(\underbrace{\sin x}_{1})$ $\quad (\infty \times 0)$

$\Rightarrow \lim_{x \to \pi/2^-} \ln y = \lim_{x \to \pi/2^-} \frac{\ln(\sin x)}{\cot^2 x}$ $\quad \left(\frac{\infty}{\infty}\right)$

$= \lim_{x \to \pi/2^-} \frac{\sin^2 x \ln(\overset{1}{\sin x})}{\cos^2 x}$ $\left(\frac{0}{0}\right)$

$= \lim_{x \to \pi/2^-} \frac{2\sin x \cos x \cdot \ln(\sin x) + \sin^{\cancel{2}} x \frac{\cos x}{\cancel{\sin x}}}{-2\cos x \sin x}$

$= \lim_{x \to \pi/2^-} \frac{\cancel{\sin x}\cancel{\cos x}\left[2\ln(\sin x) + 1\right]}{-2\cancel{\sin x}\cancel{\cos x}}$

$= -\frac{1}{2}$

$\Rightarrow \lim_{x \to \frac{\pi}{2}^-} y = e^{-1/2}$